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Training a differentiable conceptual functional equivalent of the US national water model to estimate parameters for use in NextGen

Ziyu Li¹, Andy Wood^{2,3}, Daniel McKenzie¹, and Jonathan M. Frame⁴

¹Applied Mathematics and Statistics, Colorado School of Mines, Golden, CO, United States of America

²Hydrologic Science and Engineering, Colorado School of Mines, Golden, CO, United States of America

³NSF National Center for Atmospheric Research, Boulder, CO, United States of America

⁴Department of Geological Sciences, University of Alabama, Tuscaloosa, AL, United States of America

Hydrologic model calibration can be challenging even with sufficient observations to constrain local model parameters, and far more difficult when estimating parameters across large domains – a process known as parameter regionalization. In recent years, the use of machine learning in differentiable hydrologic modeling has shown potential to address this regionalization problem. Here, a neural network learns to predict model parameters from meteorological forcings and geophysical catchment attributes by updating its weights using gradient-based optimization to minimize a loss function that quantifies the discrepancy between the conceptual model's simulations and the observations. Such a model trained over a large set of basins at once will learn regional hydrological behaviors and can be used for parameter regionalization. We investigate whether this approach can be used to determine static parameters for NOAA's Next Generation Water Resources Modeling Framework (NextGen), specifically for the National Water Model Conceptual Functional Equivalent (CFE) model by embedding a differentiable version (dCFE) into the NeuralHydrology (NH) platform for training and extracting the trained neural network to use in CFE parameter regionalization across CONUS. We introduce two ways of extracting static parameters from the neural network, and compare these to dynamic parameters obtained using the same workflow. This presentation describes this effort, including the validation of NH-dCFE to dCFE and CFE, successes in three modes of training, and the challenges encountered. We also offer recommendations on strategies to advance this parameter estimation approach in the future.